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Status of Thesis

Note: red = pre first edit, orange = pre second edit and beyond, green = done, blue = may not make it into the thesis, or it may be moved to the appendix.

- Introduction Chapter (??): Not started. Will write last/almost last
- IFO Chapter (??): Apparently this is a good chapter...
- PSL Chapter (??): Passed first drafting. Very Minor corrections to be made.
- BHD Chapter (??): Still to apply Ken and Bryan's edits (see email from 29/09/2020 'quick question' for notes).
- BHD Chapter (1): Still to apply Ken and Bryan's comments.
- SRM beam parameter Chapter (??): Still to apply Ken and Bryan's comments
- MZ chapter Chapter (??): Not written, vaguely planned
- Photodiode Chapter (??): Need to finish adding references.
- Conclusion Chapter (??): Not started. Will write last/almost last

Chapter 1

Design of the LIGO A+ HAM6

Layout

The LIGO A+ HAM6 vacuum chamber will contain the readout photodiodes, the output mode cleaners (OMC)¹ and their associated steering optics. See Figure ?? for HAM6's location with respect to the rest of the detector. Fundamentally, the optical layout in HAM6 must combine the signal beam and the LO beam and sense the combined beams to form a balanced homodyne detector (BHD). The implementation of BHD for LIGO A+ requires a complete redesign of the HAM6 layout. As losses in the interferometer's readout will degrade its sensitivity, it is vital that the active wavefront control (AWC) is designed in such a way as to minimise this.

In Section 1.1, the optical properties of the proposed HAM6 layout are examined. Section 1.2 is primarily a discussion of the design challenges associated AWC in HAM6. Section 1.3 is an analysis of how mode mismatches between the interferom-

¹The OMC assembly includes the readout photodiodes.

eter and the OMC can be corrected by the AWC. Section 1.4 is the conclusion for this chapter. This work was undertaken by a team made up of myself, Ken Strain, Russell Jones, Stephen Webster, Mark Barton and Eric Oelker.

1.1 The Chosen HAM6 Optical Layout

The design that is being taken forward at the time of writing (January 2021) is shown in Figure 1.1. Optically, there are many layouts which could have satisfied the requirements and constraints laid out in Section 1.2 . The choice of layout for HAM6 was ultimately determined by mechanical factors such as table space, number of suspensions and the preferred design for the Balanced Homodyne Single Suspension (BHSS). The optical properties of this layout are summarised in Table 1.1. The mirrors OMA1, OMB1, OMA2, OMB2 etc. will be referred to as OMx1 and OMx2 as it is not important to make a distinction between the A and B paths when considering mode-matching.

The beam size and Gouy phase as a function of position along the optical axis is shown in Figure 1.3. Significant points along the optical axis are also shown in this figure. The size of the beam at the final mode-matching optic matches the size of the target mode, as the size of the beam cannot be changed at a mirror's surface. The beam is always less than 1.2mm within HAM6, an important fact to remember when considering how close structures can be to the beam path ².

²The minimise scattering, the clear aperture size for the beam is defined to be 10 times larger than the Gaussian beam width.

As astigmatism degrades the mode-matching, it can contribute to the loss in the readout of the detector. Non-zero angles of incidence (AOI) of non-astigmatic beams onto curved optics cause the reflected beam to become astigmatic (for a comprehensive description of astigmatism see [1]). For this layout, the astigmatic loss was calculated to be $< 0.05\%$; see Appendix B for detail in this type of calculation. The astigmatic loss is plotted as a function of angle in Figure 1.2; as the beam will have low curvature, the loss becomes significant when the AOI is above 10° . The compromise that is made when choosing a larger AOI is between the potential for larger Gouy phase separation between OMx1 and OMx2 at the expense of loss due to astigmatism. In reality the IFO and OMC will also have a small amount astigmatism so this calculation will not be entirely correct; however, as some of the losses in the readout train of the interferometer are expected to be at the many percent level, losses of $< 0.05\%$ are considered unimportant.

The focal lengths of the OFI lens, OMx1 and OMx2 were optimised so that there was as close to 45° Gouy phase separation between OMx1 and OMx2 as possible as this gives orthogonal control over the mode-matching between the arm mode and the OMC. This is shown in Figure 1.4. With the layout shown in Figure 1.1, it is possible to achieve 42° Gouy phase separation between OMx1 and OMx2. Other layouts could provide exactly 45° Gouy phase separation; however, this layout offers many practical benefits, mainly in relation to the BHSS, over the other considered designs that outweigh an additional 3° of Gouy phase separation between OMx1 and OMx2. This layout also provides good Gouy phase separation for alignment control as there is $> 80^\circ$ between OMxS and OMx2.

Optic	Power (1/m)	Focal length (m)	RoC (m)
OFI lens	-0.066	-15.068	-
OMx1	-0.350	-2.855	5.709 (concave)
OMx2	-0.847	-1.180	2.360 (concave)

Space Name	Optical Path Length (mm)
SRM to OFI lens	1120
OFI lens to HAM6 table edge	720
HAM6 table edge to OM0	500
OM0 to BHDBS2	946
BHDBS2 to OMxS	Path A: 225, Path B: 315
OMxS to OMx1	Path A: 1240, Path B: 1150
OMx1 to OMx2	1590
OMx2 to OMCx waist	1220

Parameter	Value
SRM Complex Beam Parameter	$q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$
OMC waist	$485 \mu\text{m}$
OMx1 to OMx2 Gouy phase	42°
OMxS to OMx2 Gouy phase	Path A: 85° , Path B: 81°
OMx1 and OMx2 AOI	A: 9.7° , B: 4.9°

Table 1.1: Mechanical and optical properties of the LIGO A+ HAM6 layout.

In conclusion, the proposed HAM6 layout, shown in Figure 1.1, fulfils the criteria set in Section 1.2. Near optimal mode-matching and alignment control can be achieved with this layout, and a summary of the properties of the layout can be found in Table 1.1. The next stage of development for this layout involves the design and prototyping of parts and the consideration of fine details such as cable routing.

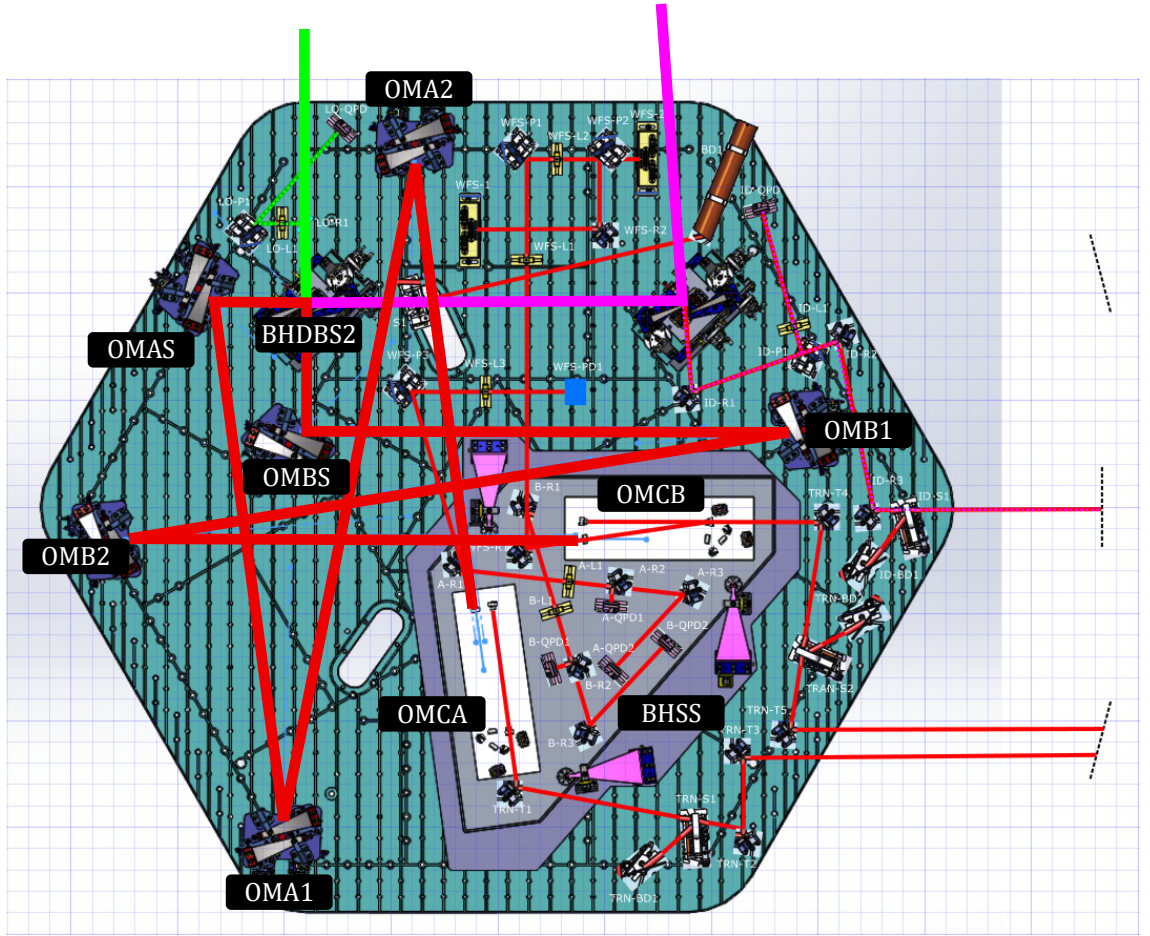


Figure 1.1: Place holder graphic. Caption to the effect of... The LO beam enters HAM6 directly above BHDBS2 (north side), and it is highlighted in green. A pick-off will be used to monitor the alignment of the LO; the pick-off beam is indicated with a dashed green line. The signal beam enters HAM6 directly above OM0 (north side), and it is highlighted in pink. The signal beam is steered by OM0 onto BHDBS2 where it is combined with the local oscillator. The shutter is located between OM0 and BHDBS2. A small amount of the signal beam is transmitted by OM0 (dashed pink); this beam is used align ???. The two combined beams are shown in red. These are incident upon a mirror used for steering (OMxS) and then upon two mirrors curved mirrors (OMx1 and OMx2). (Caption continued on next page).

Figure 1.1: (Figure on previous page). The BHSS (the large triangular component) takes up most of the bottom left of the HAM6 chamber. The position of the OMCs is shown. On the BHSS, there are several small optics used to steer beams that are required to control ??; the wavefront sensors (yellow components) for this are found at the top of the HAM6 table. Several beams are directed of the table for imaging with CCD cameras.

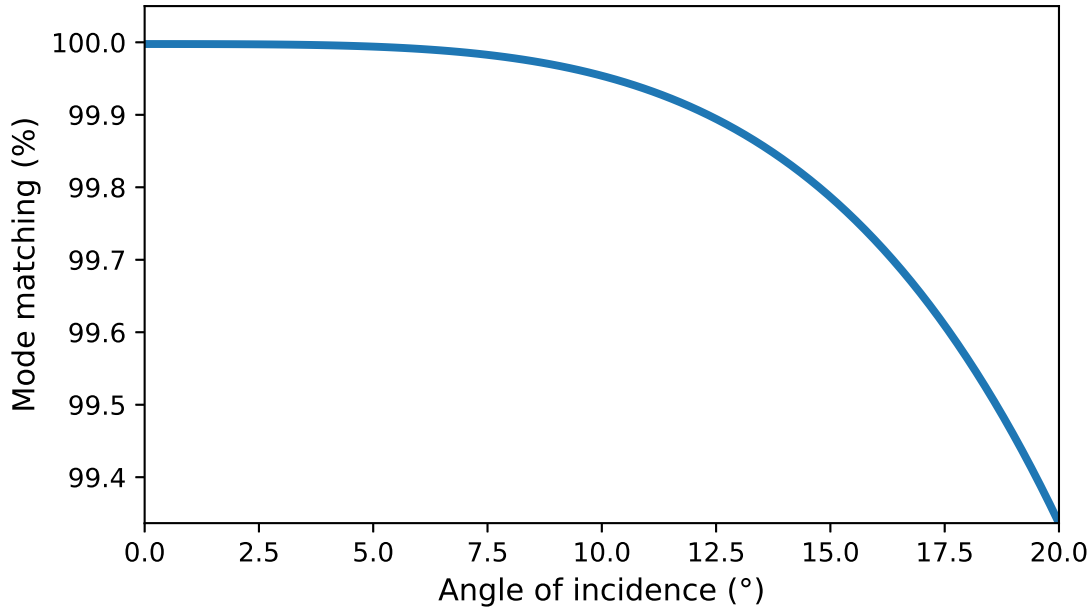


Figure 1.2: Astigmatism is a source of loss. When designing the layout shown in Figure 1.1, the AOI for the curved mirrors was altered to explore how sensitive it was to astigmatism. Using Equation (1.2), the loss was computed. This effect is negligible up to 10° after which it starts to become significant compared to the losses summarised in Table 1.2.

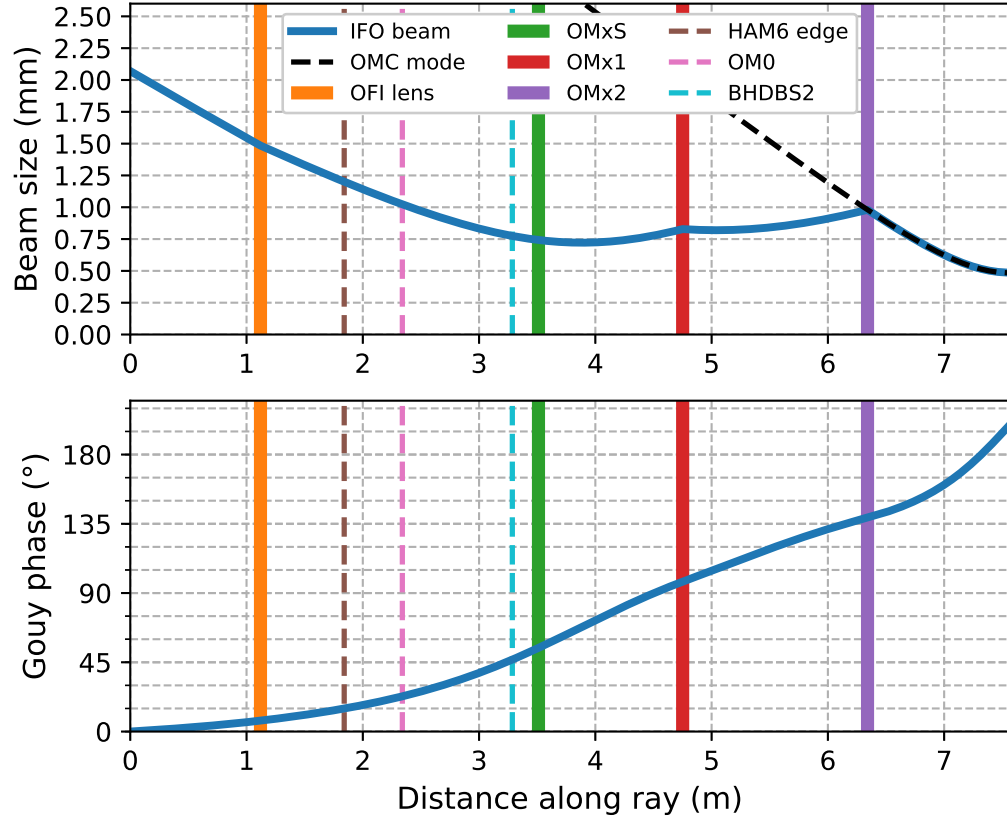


Figure 1.3: The top panel shows the radius of the beam as it travels from the SRM ($z = 0$) to the waist of the OMC. Significant points along the axis are marked with vertical lines. Less significant, but useful, reference points are marked by dashed vertical lines. The bottom panel shows the Gouy phase shift accumulated by the beam as it propagates along the optical axis.

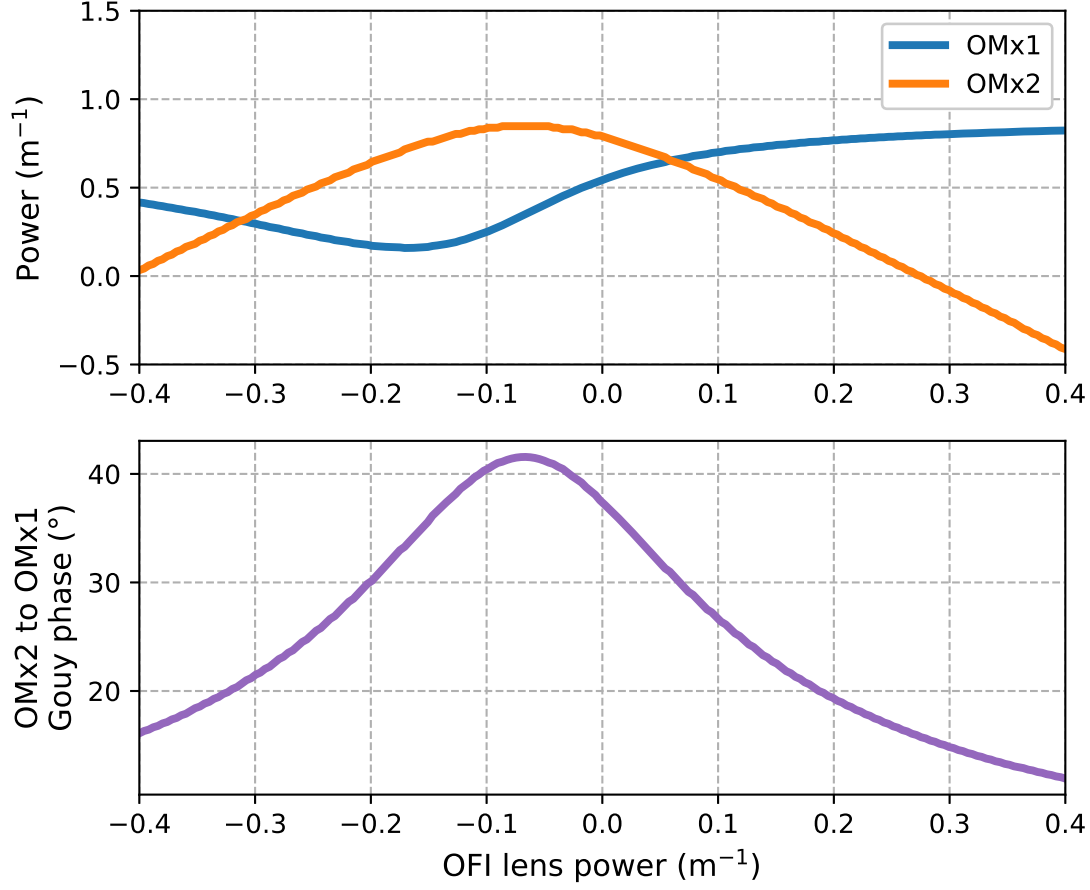


Figure 1.4: Top panel: The choice of OFI lens power affects the combination of OMx1 and OMx2 power that is required for 100% mode-matching. Bottom panel: The best choice for of OFI lens that corresponds to obtaining orthogonal control of the beam parameter in HAM6 is where the Gouy phase separation between the pair of OMx1 and OMx2 is closest to 45° . The most Gouy phase separation between OMx1 and OMx2 achievable by this layout is 42° .

1.2 The Requirements for the Optical Layout Within HAM6

The BHD optical layout in HAM6 is required to provide a readout of the gravitational wave signal. To meet the design sensitivity of A+, HAM6 must house a layout that is robust to mode mismatch as there is significant uncertainty in the beam parameter of the arm cavity mode at the SRM. Having control over the two degrees of freedom associated with mode-matching – the size and location of the beam’s waist – between the OMC and the beam exiting the SRM was challenging as this requires a lot of space relative to the size of the optical table. In addition to this, the HAM6 layout needs to leave space for the other sensors and beams used for controlling the interferometer. All proposed designs used most of the table space. This section focusses on the design considerations relating to mode-matching; other aspects of the functionality required from HAM6 is summarised in [2].

To reach the desired level of squeezing for A+, the goal for mode-matching between the arm mode and the OMC mode is $> 98\%$; the loss contribution of the components in the readout train of the interferometer are summarised in Table 1.2 [3]. Active mirrors, called Suspended Active Matching Stages (SAMS), will be used to correct for mode-mismatch; they will provide at least ± 50 mD of optical power tuning range [4].

The AWC in HAM6 has two functions: it will be used to correct for thermal effects in other parts of the interferometer [2] such that the mode-matching between the arm mode and the OMC is maximised, and it will be used to correct for RoC

errors in the interferometer after the main beamsplitter.

Thermal changes in the arm cavity mode may be corrected for by the thermal compensation system (TCS) [5]. However, in practice operators and commissioners use each tunable aspect of the interferometer to improve its sensitivity to gravitational waves. For instance, at LLO it has been found that TCS can be used to decrease laser intensity noise coupling to DARM [6]. As each interferometer subsystem may not be used for its intended purpose, having redundancy in the design of the interferometer is desirable if unexpected noise couplings are discovered. As TCS is used to decrease a noise coupling, it may not be able to correct thermally-driven errors in the arm mode. The optical layout and AWC functionality in HAM6 were designed with this flexibility in mind.

Two curved mirrors (OMx1 and OMx2) are required to control the two degrees of freedom associated with mode-matching. To gain flexibility in the design, a lens will be used to provide further Gouy phase separation between the two active mode-matching mirrors in each path after BHDBS2. There is space for this lens on the OFI assembly; this is vibrationally isolated and is a useful distance from the mode-matching optics in HAM6. An active lens was considered here, but concerns of OFI assembly heating [7] and the unavailability of suitable flint glass meant that this idea was discarded.

To provide orthogonal mode-matching control, the two SAMS need to be separated by 45° of Gouy phase. Due to the design of the OMC, the Rayleigh range of the beams in HAM6 are of the order of one meter. As most of the Gouy phase is accumulated within one Rayleigh range either side of a beam waist, a beam will

have to travel a significant distance across the table before it acquires sufficient Gouy phase. This Gouy phase separation requirement gave us a method of generating candidate designs as this requirement can be interpreted as needing the mode-matching mirrors to be as far apart as possible from each other on the HAM table. Losses due to astigmatism prevents one from using a large AOI to increase the separation between optics in HAM6.

Orthogonal alignment control is obtained by having two mirrors separated by 90° of Gouy phase. For all designs, the two mirrors for alignment were immediately after BHDBS2 and immediately before the OMC as this gave the best Gouy phase separation between them. It was found that placing the first mirror close to BHDBS2 often provided good layouts. References [8, 9] describes how the longitudinal and pointing degrees of freedom associated with BHD may be sensed and controlled.

The OMCs are expected to be the largest contributor of back-scatter from HAM6 to the interferometer [2, 10]; as described in [10], backscatter can degrade the sensitivity of the interferometer. A single stage suspended platform (BHSS) is required to prevent backscatter noise from reducing the interferometer’s sensitivity. Both OMCs will need to be on this platform and so all designs need to have the OMCs fairly close to each other. A large BHSS is mechanically infeasible as it will reduce space for the other components on the HAM6 table.

1.2.1 Fixed properties of the HAM6 Optical Layout

There are a few properties of the HAM6 layout that will not be changed. These include: the entry point of beams into the HAM6 chamber, the size of the nominal

mode entering HAM6 and the size of the OMC waist. Table extensions can be made in places where they would not impede access, however the size of the HAM optical table does restrict the design. It was decided that the two beam paths after BHDBS2, the optic on which the LO and signal is combined, would be as similar as possible as both beams need to be matched to OMCs of the same design.

The nominal beam parameter at the SRM is $q_{\text{SRM}} = -3.55 + 1.09i \text{ m}$ [11], so the design of the HAM6 mode-matching optics is based on this quantity. As discussed in Chapter ??, the SRM beam parameter is expected to differ from its nominal value. It is described in Chapter ?? how geometry errors in the SRM cause the arm cavity and the SRC to be mismatched. This means that the radius of curvature and beam size at the SRM will not correspond to the eigenmode of the SRM, and the beam parameter at the SRM that should be considered corresponds to the arm cavity mode. This set of likely modes, shown in Figure ??, was taken into consideration when choosing a layout for HAM6 since different candidate layouts may have been more or less robust to beam parameter errors at the SRM.

The OMC design developed for aLIGO [10], with trivial modifications, will be used to remove junk light from the signal beam and so there is no choice in target waist size. The nominal OMC waist size is $485 \mu\text{m}$. The beam enters the OMC through a flat mirror, and a waist is located 140 mm from this mirror. There is a negligible amount of astigmatism inherent in the design of the OMC. Mode-matching optics therefore have to match the nominal SRM and OMC modes.

HAM6 loss Source	Target for O5
OMC throughput loss	< 2%
OM3 transmission (or equivalent) for sensing	1–2%
PD quantum efficiency	1%
Mode mismatch to the OMC	< 2% (as low as possible)
Any other optical loss	< 1%

Table 1.2: Target for losses in HAM6. Reproduction of [3]

1.3 Visualisations of the Robustness of the A+ HAM6

Layout to Mode Mismatch

As described in Section 1.2 and Chapter ??, it is expected that there will be some mismatch between the arm mode and the OMC. To visualise the effectiveness of the AWC provided by the layout’s active optics, it is useful to consider a 2D space representing Gaussian beams. The region of this space that can be mode-matched from by the AWC and a distribution of modes that those optics may have to correct for can be compared to gain insight into the effectiveness of the layout.

The proposed layout in Section 1.1 can mode-match the majority of beam parameters one would expect based on uncertainties in the RoC of the optics within the SRC. Despite this good coverage, measurements from LLO show that the beam parameter may fall outside the region of WS which the layout can compensate for as the LLO SRC differs significantly from the nominal design. Alternative optical powers for the OFI lens, OMx1 and OMx2 are given in Section 1.3.5; this change would sacrifice orthogonality of the mode-matching and alignment optics in favour of robustness to RoC errors in the SRC.

1.3.1 A Description of Mode-Matching in Terms of HG Modes

The mode-matching between two laser beam modes, or a laser mode and a cavity mode, describes how similar those two modes are. In general, a combination of misalignment and differing beam shape can be considered to be a mode-mismatch. However, mode-matching typically refers to the overlap of the two modes if they were co-aligned, thus it is a measure of how similar in shape the two modes are. Mode-matching between two modes is usually, to first approximation, thought of as a difference in their waist size and waist location.

Small mode-mismatches can be modelled as light scattering from the beam's, or cavity's, fundamental mode into its $\text{HG}_{20} + \text{HG}_{02}$ modes. If a cavity is designed such that the fundamental mode will be resonant while the $\text{HG}_{20} + \text{HG}_{02}$ modes will be reflected, the effect of mode-mismatch between the cavity and a signal carrying beam is to decrease the signal light that is transmitted by the cavity since some of the signal carrying light will be scattered into the reflected $\text{HG}_{20} + \text{HG}_{02}$ modes. Sensing the $\text{HG}_{20} + \text{HG}_{02}$ modes therefore gives a measure of the mode-matching between two modes.

A change in waist size is equivalent to an in-phase coupling between the fundamental mode and the $\text{HG}_{20} + \text{HG}_{02}$ modes, whereas a change in waist location is a coupling to the $\text{HG}_{20} + \text{HG}_{02}$ modes which is 90° out of phase with the fundamental mode [12]. As the $\text{HG}_{20} + \text{HG}_{02}$ modes acquire 2ϕ compared to the fundamental mode, where ϕ is the Gouy phase acquired by the fundamental mode, spacing beam curvature actuators at 45° will give an orthogonal basis for control over mode-matching. Orthogonal control results in the beam size and beam waist, or two

combinations of them that are orthogonal, can be independently controlled.

1.3.2 Calculating Mode-Matching Using The Gaussian Beam Parameter

Two perfectly mode-matched beams will have identical beam parameters at the same point along the optical axis. The overlap integral between two fields, E_1 and E_2 ,

$$\mathcal{M} = \frac{\left| \int_A E_1^* E_2 dA \right|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}, \quad (1.1)$$

can be used to determine the amount of mode-matching, $\mathcal{M}$, between two astigmatic modes. Solving this integral gives

$$\mathcal{M} = \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}, \quad (1.2)$$

where $q_{1,2}$ is the beam parameter of the electric field $E_{1,2}$. This is derived in Appendix B. The solution to Equation (1.2) is always $0 < \mathcal{M} \leq 1$. Alternatively, mode-matching may be computed in terms of the amplitude of the higher order modes relative to the fundamental mode [13]. The mode-matching can be calculated at any point along the optical axis of the layout as mode-matching must be preserved throughout it; however, it makes sense to calculate the mode-matching in a physically significant plane such as the input plane, a plane containing a curved optic or the plane containing the waist of the target beam.

1.3.3 WS Space Visualisations

A Gaussian beam can be described by its beam size, w , and defocus, S [14]. A WS space plots a Gaussian beam in terms of these two parameters. The effect of focussing elements on Gaussian beams can be clearly seen in this representation of them. Because a curved mirror can only change the curvature of a beam, at the mirror's surface the beams before and after reflection must have the same size. The process of a Gaussian beam, described by $q_1(S_1, w_1)$, being reflected from a mirror with power S_m to produce a new Gaussian beam, q_2 , can be written as

$$\frac{1}{q_2} = S_m + \frac{1}{q_1} = (S_m + S_1) - i\frac{\lambda}{\pi w_1^2}. \quad (1.3)$$

From Equation (1.3) it can be seen that a mode moves in the S direction when it reflects from a curved mirror. This representation of Gaussian beams has the advantage of showing the beam in terms of intuitive quantities. The downside to this representation is as the beam propagates through the layout, the path that is traced through the WS space by the mode is fairly complex.

There are two other ways that a beam parameter could be graphically represented. Since q is a complex number, it could be represented in terms of its real and imaginary components. This makes seeing the effect of propagation straight forward, but the effect of a focussing element is harder to visualise. Alternatively, plotting $1/q$ in terms of $\lambda/\pi w^2$ and S produces plots where the effect of focussing elements and propagation are easy to visualise [15, p.680]. Curved mirrors and lenses result in a change in defocus, and propagated modes follow circular paths. As these figures are being used to visualise mode-matching and the effect of actuators on the beam size

(rather than $\lambda/\pi w^2$) and defocus, the WS representation was chosen. One could express the beam parameter in terms of natural units by dividing the beam size by w_0 and the defocus by $1/z_R$.

To illustrate how actuators move a mode in WS space, consider two lenses with variable power separated by 45° of Gouy phase. As the two lenses are orthogonal, the waist size and waist location of the mode exiting this layout will be orthogonally controllable. In a WS space, the orthogonality of two actuators can be seen by the angle between the two vectors that represent each actuation. If the two lenses are orthogonal then the actuation vectors will be orthogonal. Figure 1.5 is a sketch of the WS space representation for a simple optical layout consisting of two active lenses separated by 45° Gouy phase.

The mode-matching requirement between the interferometer and the OMC for LIGO A+ is $> 98\%$. It is useful to plot contours of the mode overlap in the WS space between a mode and a target mode. Note that these contours can only tell you about the overlap between a mode and the target mode; the mode-matching between two arbitrary modes can not be deduced from these contours [14]. For example, two modes may be mismatched by the same amount from the target in orthogonal directions in WS space. They cannot be 100% mode-matched as these two modes are different from each other.

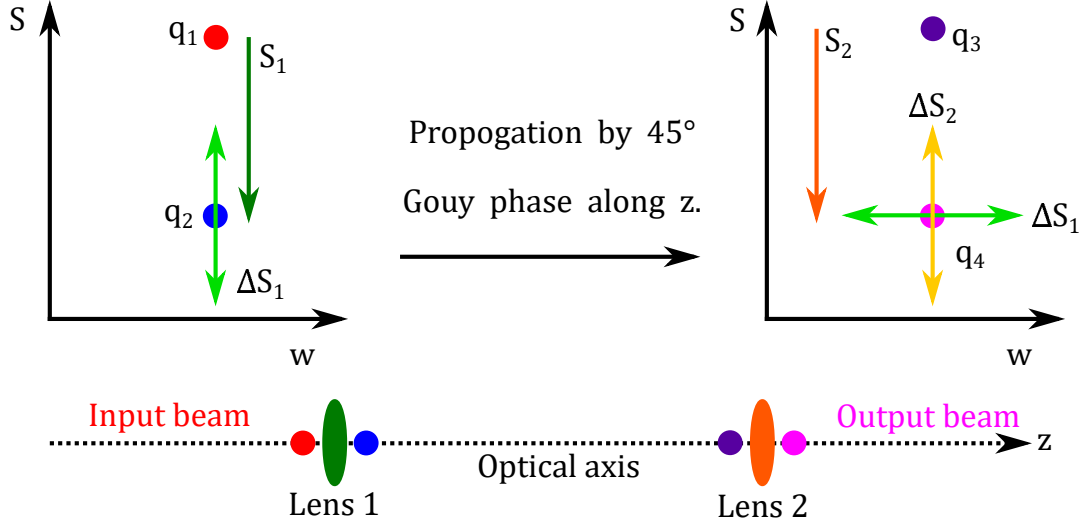


Figure 1.5: Sketches of the WS representation for a Gaussian beam interacting with different parts of a simple optical layout consisting of two active lenses with powers $S_{1,2} \pm \Delta S_{1,2}$. The left hand axes shows the WS plane at Lens 1. The input beam, q_1 , is indicated by a red dot, and the effect of the lens on q_1 is shown by the dark green arrow S_1 . This results in a new mode q_2 which is shown as a blue dot. A light green arrow indicates the actuation of Lens 1. The right hand axes shows the WS plane at Lens 2. The input mode at Lens 2, q_3 , is shown as a purple dot, and the output mode, q_4 , is shown as a pink dot. The static and dynamic power of Lens 2 is indicated by the orange and yellow lines. As Lens 1 and Lens 2 are separated by 45° of Gouy phase, the actuation provided by Lens 1 results in a change in the beam parameter which is orthogonal to the actuation provided by Lens 2, provided that the actuations are small enough. Beneath the two WS space diagrams is a sketch of the optical axis and the location of the modes.

1.3.4 Visualising how Active Mode-Matching Optics Expand the Area of WS Space Representing Acceptable Input Modes

To visualise the effectiveness of the actuation provided by the active elements of the HAM6 layout, the region of the WS space at the SRM from which the target mode-matching space can be achieved was computed. To do this, first the ring of modes in WS space that is 98% mode-matched to the OMC after reflection from the second active mirror, OMx2, was considered. If OMx2 had a static radius of curvature then the effect of OMx2 would be to move the ring vertically in the WS plane. However, the ring can also be stretched in the S direction as OMx2's radius of curvature can be changed. This can be seen, crudely, by applying $S_{\text{OMx2}} \pm \Delta S$ to the original $> 98\%$ mode-matching ring to produce two new rings separated by $2\Delta S$.

To generate the region from which the target region can be reached by OMx2, the target region should be cut by a line which intersects the ring at the extrema of the beam size. To maximise the amount of WS space covered by the actuator, the upper target region should have $+\Delta S$ applied to it whereas for the lower region the actuation should be $-\Delta S$. This results in a region of WS space which is larger than the initial target region. All modes in the new region can be at least 98% mode-matched to the OMC. The new region was then propagated to the first mirror, OMx1, where the same procedure can be applied to further expand the region. This process is shown in Figure 1.6.

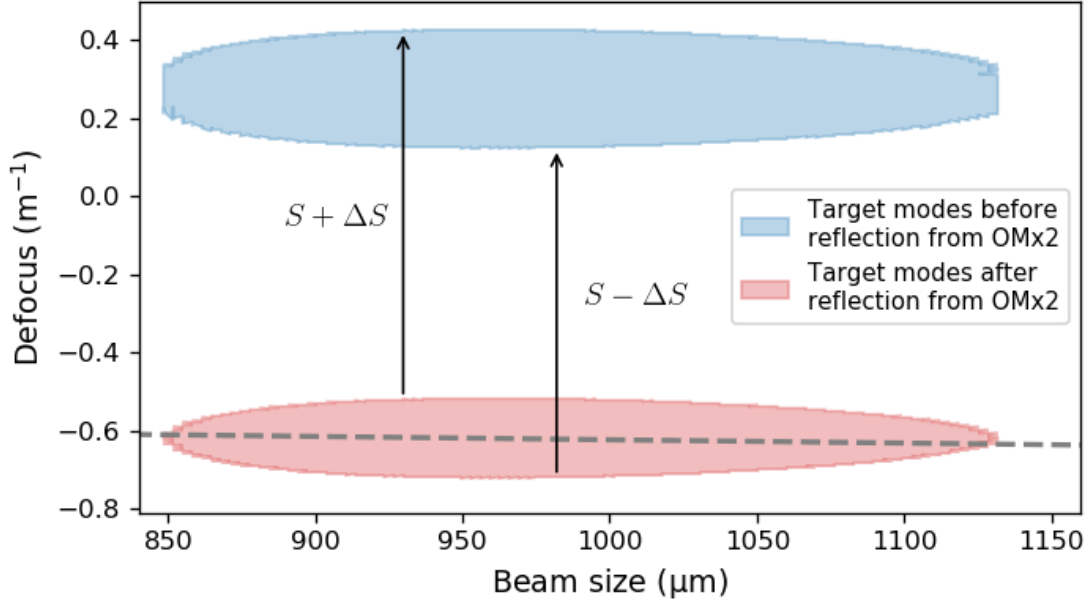


Figure 1.6: The WS space at the OMx2 plane. The effect of OMx2 in the plane of OMx2 is to move a mode vertically in WS space. After reflection from OMx2, a mode must be at least 98% mode-matched to the OMC as there are no further active mode-matching elements in the layout of HAM6. Modes with at least 98% mode-matching are shown by the red region. The blue region represents modes which can be reached from the red region by either $S_{\text{OMx2}} \pm \Delta S$, and was constructed by applying transformations to the modes in the red region. The red region was split into two parts, ‘upper’ and ‘lower’, indicated by the dashed line that connects the two extremes in width of the region. To create the lower edge of the blue region, modes in the lower edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} - \Delta S$ applied to them. To construct the upper edge of the blue region, the upper edge of the red region had the transformation corresponding to a mirror power of $S_{\text{OMx2}} + \Delta S$ applied to them.

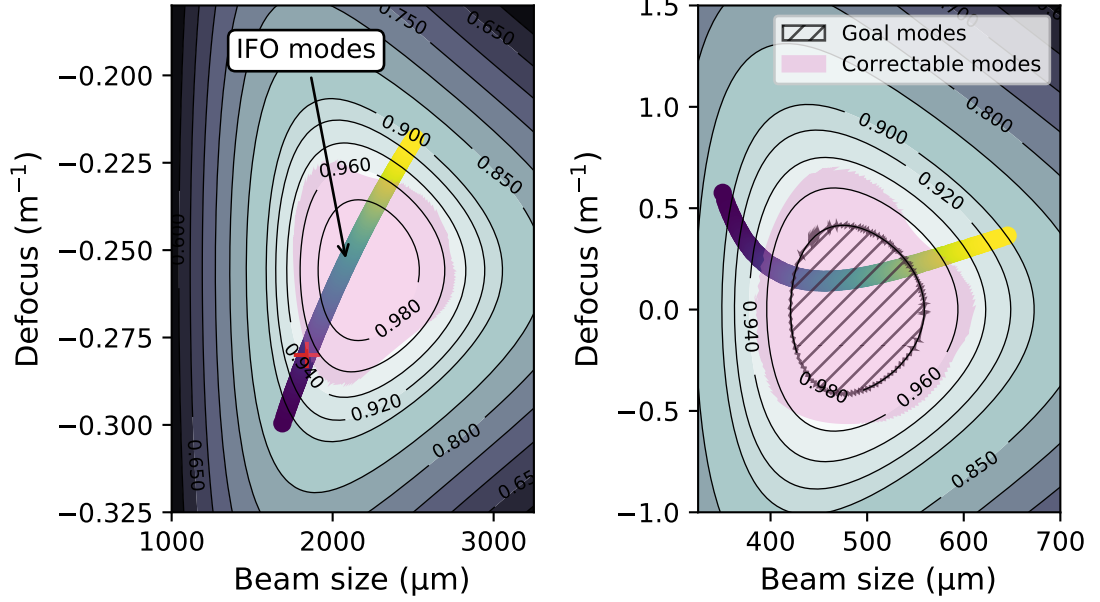


Figure 1.7: The region of WS space covered by the AWC in HAM6 if OMx1 and OMx2 for the layout discussed in Section 1.1. Left panel: the set of modes that is expected to be present at the SRM, as described in Chapter ??, is shown by the purple-blue-green-yellow curve. The pink region shows which modes that can be corrected for so that 98% mode-matching is achieved. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. The red cross indicates the mode size that can be inferred from measurements of the SRC's Gouy phase at LLO. Right panel: this shows the same modes as the left panel, except they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region, which is indicated with a hatched area.

The modes from the SRM that can be matched to the target are shown in Figure 1.7. To visualise how successful the layout is at handling the likely modes from the interferometer (this set of modes was investigated in Chapter ??), the interferometer modes were compared to the region of WS space containing modes which can be corrected such that they meet the $> 98\%$ mode-matching requirement. It was found that majority of likely modes will be covered by the layout as the modes near the edge of the interferometer mode continuum are more unlikely to occur than the ones near the centre; however, measurements of the SRC's Gouy phase at LLO can be used to infer the beam parameter at the SRM, and this mode would fall outside of the region of WS space which can be compensated for by the AWC. Its worth noting that the modes at the yellow end of the continuum shown in Figure 1.7 may not be physical as they correspond to an unstable SRC.

The shape of the pink region in Figure 1.7 is nominal; when the HAM6 layout is constructed the pink region's shape will be altered by geometric errors in OMx1 and OMx2. Nevertheless, Figure 1.7 is encouraging since it shows that the actuation provided by OMx1 and OMx2 is highly tolerant to mode-matching errors generated by errors in the SRC.

1.3.5 Optimising WS Space Coverage in Favour of Expected Mode Errors

While 45° of Gouy phase separation between the mode-matching optics in HAM6 is a desirable quality for the layout to have, with an actuation range ± 50 mD, the layout can be optimised to correct for mode-mismatches between the OMC and the continuum of expected beams exiting the interferometer. The majority of the

uncertainty in the interferometer's beam parameter comes from the uncertainty in SR3's RoC (see Figures ?? and ??). Ideally, one would correct for this by actuating on the RoC of SR3, however the RoC of SR3 cannot be strongly actuated on. If a mirror was placed 180° away from SR3, it could be used to compensate for the uncertainty in the RoC of SR3 i.e. one could configure the mode-matching optics within HAM6 such that they are biased for correcting the errors in the interferometer beam parameter caused by the error in SR3.

By reducing the orthogonality of the AWC actuators, more range can be obtained for one of the beam's property e.g. the size of a beam's waist. A Gouy phase separation ($\sim 30^\circ$) offers a compromise between orthogonality versus actuation range for errors in one of the beam parameters dimensions. Figure 1.8 shows how the region of WS space covered by the HAM6 AWC can be biased in favour of the expected set of modes that may emerge from the SRM. The physical positions of the optics are the same as before (see Table 1.1), however the optical powers of the OFI lens, OMx1 and OMx2 was altered (-0.2, 1.67, 0.631 respectively) to change the Gouy phase separation between OMx1 and OMx2 to 30° . However, the Gouy phase for alignment control would be significantly reduced; this is shown in Figure 1.9.

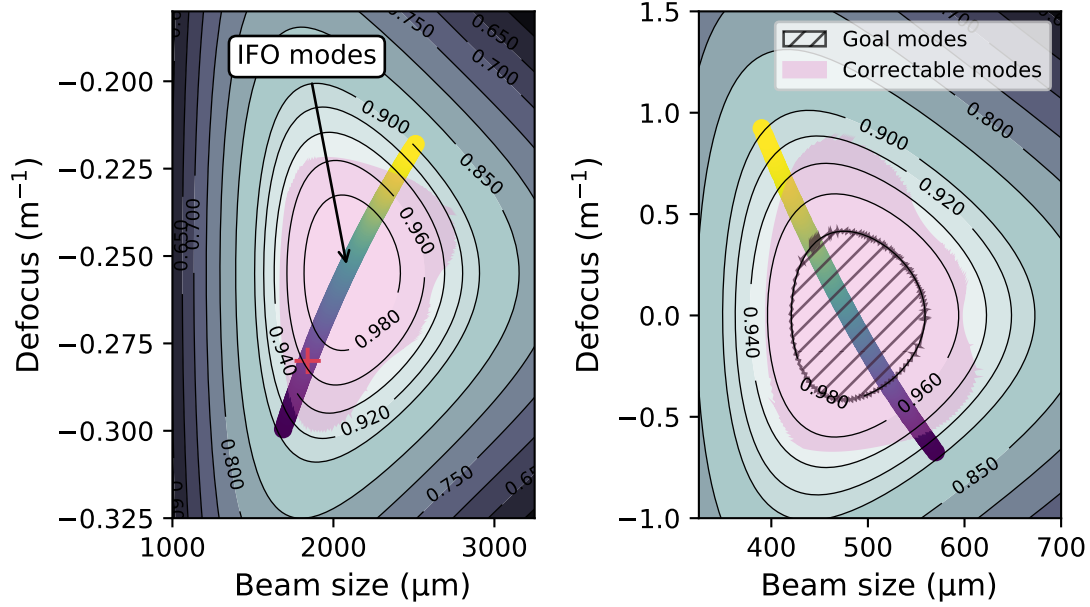


Figure 1.8: At the expense of orthogonality, the region of WS space covered by the AWC in HAM6 can be optimised by changing the Gouy phase separation between OMx1 and OMx2. Left panel: the set of modes that could be emerge from at the SRM, as described in Chapter ??, is shown by the purple-blue-green-yellow curve. The pink region shows which modes can reach 98% mode-matching by using the AWC in HAM6. The contours indicate the mode-matching between a mode (W, S) and the OMC mode. Right panel: this shows the same modes as the left panel; however, they have been transformed by the optical layout to the plane of the OMC waist. Modes within the pink region can be actuated on so they move to the goal region (hatched area).

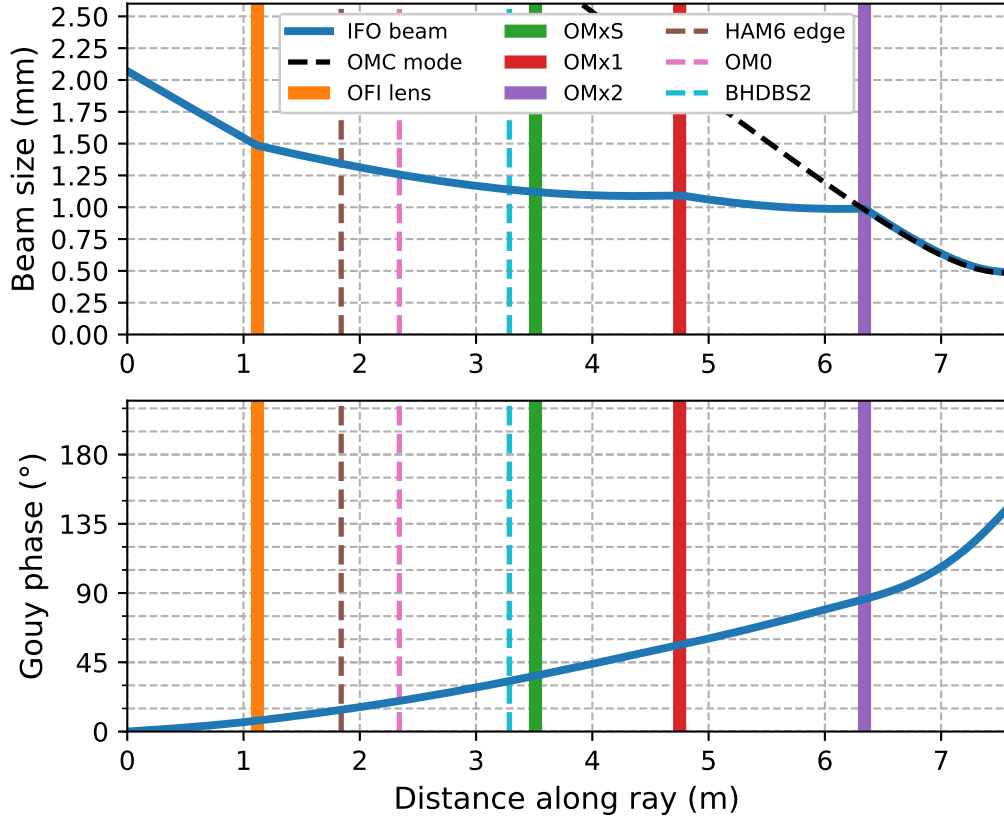


Figure 1.9: Top panel: the beam profile for the alternative layout of HAM6 suggested in Section 1.3.5. Bottom panel: The Gouy phase accumulated along the optical axis for the alternative layout. While alignment Gouy phase is degraded, the mode-matching Gouy phase was changed to bias the optical layout for correcting errors in the RoC of SR3.

1.4 Conclusion

The functionality required from the BHD layout was discussed, and a proposed layout that satisfies these requirements was presented in Figure 1.1. The optical properties of this layout were investigated. By selection of the right combination of mode-matching optics, near orthogonal control of the signal-LO mode-matching and beam alignment to the OMC was achieved.

The effect that the AWC had on the region of WS space that contains modes which satisfy the $> 98\%$ mode-matching requirement space was investigated. WS diagrams were used to represent how robust the layout is to initial mode-mismatch. Figure 1.7 shows the amount of WS space that the AWC could correct for. It would be desirable to shape this region to improve the chances of being able to mode-match the interferometer to the OMC. At the expense of AWC and alignment orthogonality, the mode-matching optics in HAM6 could be changed from their nominal values to improve the likelihood that the AWC can correct for mode mismatches as described in Section 1.3.5.

Appendix A

Algorithm For Finding a Three Optic Solution with 45° Gouy Phase

Separation between two of the Optics

To find a mode matching solutions between the SRM and the OMCs which have 45° phase separation for the AWC optics, an algorithm was developed. To visualise how this algorithm works see figures A.2 and A.1. Since a beam can be described by two parameters, its defocus S and width w , the modes can be shown in a 2-D space [14] (figure A.2). The action of a lens or curved mirror is to move the mode in the S direction. It is helpful to define a mode continuum along which w and/or S varies. The principle by which this algorithm works is: The continuum of modes after optic 1 and the continuum of modes just before optic 3 can be easily obtained since they must have a defined beam size, so they only vary in defocus. These two continuum are propagated to optic 2, and then its just a case of finding how to move from one continuum to an other.

The continua at optic 2 do not necessarily have to have the shape that they are sketched with in figure A.2, however if the blue-green and red-yellow continuum extend far enough they will always wrap around. It is also not guaranteed that every mode in the red-yellow can be reached from the blue-green continuum, and vice versa. Since the mapping between the blue-green and red-yellow continuum is not one to one, there is some choice in you can make. Both ‘branches’ of the continua, where the continuum turns back on itself in the w direction, do not need to be analysed at the same time, but a better solution may be found by looking at different combinations of branches. This is useful in setting a limit on the size of the continuum you search.

In our implementation of this algorithm, the continua are actually discrete, which means that when finding a mapping at optic two it is not guaranteed that a mode in the red-yellow continuum will have a corresponding mode of the same with in the blue-green continuum. Because of this granularity, if you were to plot optimal values of optic 2 and optic 3 power as a function of optic 1 power, you would see jumps in the optimal lens choices. This is especially bad around where the turning point in the continua is, since the modes defocus changes quickly while their size does not.

A.0.1 Method

This section delineates the method described in section A and can be visualised using figure A.2.

Define the target mode at the optic 3. In figure A.2, this is mode l at optic three.

Define a set of modes which share the same beam size as the target mode at optic 3 but have varying radii of curvature. In figure A.2, this is the red-yellow continuum at optic three.

Propagate the set of modes defined at optic 3 back to the optic 2. In figure A.2, this is the red-yellow continuum at optic 2.

Define the initial beam parameter and propagate this to optic 1. In figure A.2, this is mode i at optic one.

Define a set of modes that have the same beam size as the initial beam at optic 1 but differ in defocus. In figure A.2, this is the blue-green continuum at optic one.

Propagate the modes defined at the optic 1 to optic 2. In figure A.2, this is the blue-green continuum at optic two.

Calculate the difference in optical power between mode i and a mode in the blue-green continuum (e.g. mode j). In figure A.2, mode j represents just one possible mode in this continuum. Save this power.

Find a mode in the red-yellow continuum which is closest in size to mode j, call this mode k. In figure A.2, there is another possible choice on the red-yellow continua, however only one of them needs to be considered.

Calculate the difference in optical power between mode j and mode k at optic 2. Save this power.

Calculate the difference in optical power mode k and the target mode at optic 3. Save this power.

Calculate the Gouy phase separation of optic 2 and optic 3.

Repeat for every mode in the blue-green continuum, or until a gouy phase separation of 45degree has been found.

The number of modes that are considered and the range of modes in the continua should be tailored for the optical layout that is being considered. Our problem was with metre scale spacings between optics, and with metre scale Rayleigh lengths. It was found that continua with a range of the order 1-10 dioptres were suitable. This does not mean that an adaptive optic would need this actuation range, rather it allowed for the static radii of curvature for each optic that gives optimal Gouy phase separation between the active mode matching elements to be found. Using this algorithm, we were able to rapidly develop many possible HAM6 layouts. This allows for more time to consider various engineering factors (suspension size and the like) and how to adapt the mirror strengths accordingly to accommodate for small changes in a tentative design.

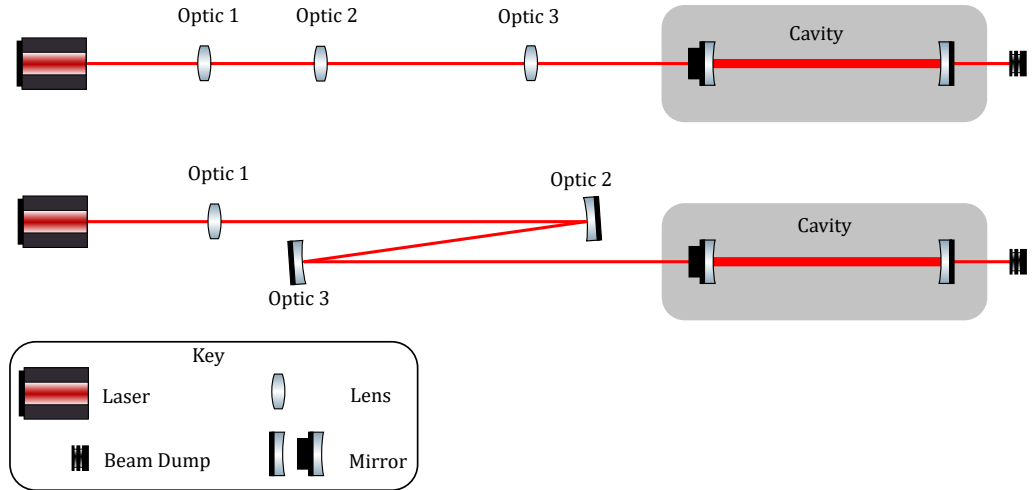


Figure A.1: Top layout: Three lenses used to achieve mode matching to the cavity. These are shown as positive lenses, however negative lenses can also be used when designing mode matching systems. Bottom Layout: A lens and two curved mirrors to achieve mode matching to a cavity. The optics 1,2 and 3 correspond to the ones shown in figure A.2. The initial and target modes are defined in these examples by the laser mode and the cavity mode.

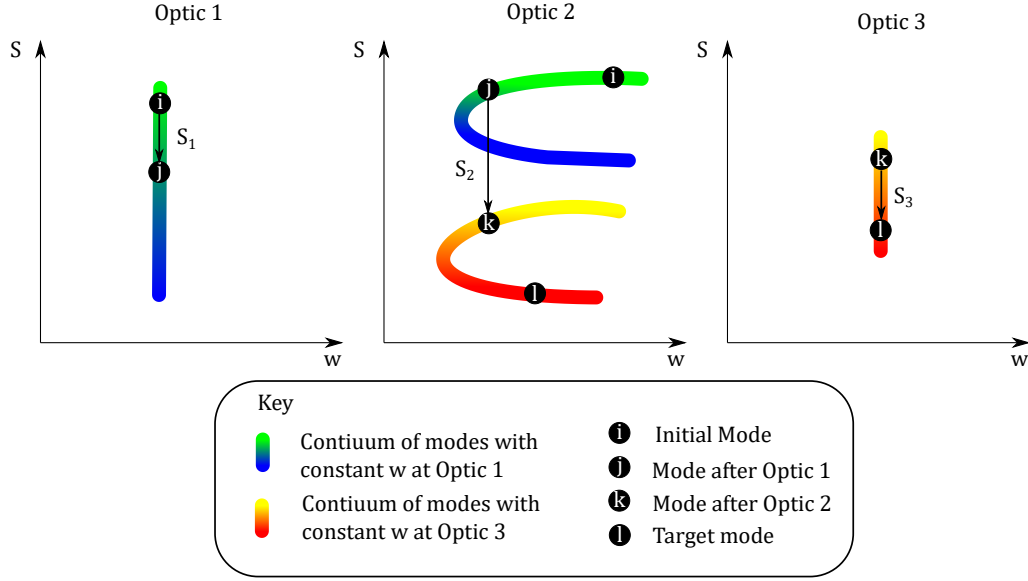


Figure A.2: Sketch of the WS phase space representations of a Gaussian beams at the three positions along the optical axis where a mirror or a lens could be. Examples of such layouts are shown in figure A.1. The optical power of the first optic is given by the difference in the defocus between the initial mode (i) and the second mode (j). The power of the second optic is found by finding the difference in defocus from the blue-green continuum to the red-yellow continuum. There are 2 optical powers that will move the mode j to the red-yellow continuum, and two possible modes in the blue-green continuum that can from which k can be reached, giving 4 possible similar solutions. The algorithm can be set up so that it is only one way of moving between the blue-green and red-yellow continua by limiting their extent. The power of the third optic is found from the difference between the mode k and mode l at optic 3.

Appendix B

Mode Matching Between 2 Astigmatic Beams

B.1 Derivation of Mode Matching TM00 Beams

The [check this naming](#) electric field, E , of a TM00 Gaussian beam can be written as

$$E(x, y, z) = E_x(x, z)E_y(y, z) \tag{B.1}$$

where

$$E_x(x, z) = \frac{1}{\sqrt{q_x(z)}} \exp\left(-ik\frac{x^2}{2q_x(z)}\right), \tag{B.2}$$

$$E_y(y, z) = \frac{1}{\sqrt{q_y(z)}} \exp\left(-ik\frac{y^2}{2q_y(z)}\right). \tag{B.3}$$

For a beam without astigmatism, $x^2 + y^2 = r^2$ and $q_x = q_y = q$ so:

$$E(r, z) = \frac{1}{q(z)} \exp\left(-ik \frac{r^2}{2q(z)}\right). \quad (\text{B.4})$$

The mode matching, $\mathcal{M}$, of two beams can be expressed by overlap of the two fields

$$\mathcal{M} = \frac{|\int_A E_1^* E_2 dA|}{\int_A |E_1|^2 dA \int_A |E_2|^2 dA}. \quad (\text{B.5})$$

The numerator of equation B.5 for TM00 beams can be written as:

$$\left| \int_A E_1^* E_2 dA \right| = \int_A E_1^* E_2 dA \int_A E_1 E_2^* dA \quad (\text{B.6})$$

$$= \int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{y1}^* E_{y2} dy \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx \int_{-\infty}^{\infty} E_{y1} E_{y2}^* dy. \quad (\text{B.7})$$

Since the calculation is the same for the x and y coordinates, we only need to show the solution for the x . The x terms of equation B.7 can be written as:

$$\int_{-\infty}^{\infty} E_{x1}^* E_{x2} dx \int_{-\infty}^{\infty} E_{x1} E_{x2}^* dx = \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}^*}} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \frac{1}{\sqrt{q_{x2}}} \exp\left(\frac{-ikx^2}{2q_{x2}}\right) dx \quad (\text{B.8})$$

$$\times \int_{-\infty}^{\infty} \frac{1}{\sqrt{q_{x1}}} \exp\left(\frac{-ikx^2}{2q_{x1}}\right) \frac{1}{\sqrt{q_{x2}^*}} \exp\left(\frac{ikx^2}{2q_{x2}^*}\right) dx \quad (\text{B.9})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*}\right) \exp\left(\frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{B.10})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2}{2q_{x1}^*} - \frac{-ikx^2}{2q_{x2}^*}\right) dx \times c.c. \right) \quad (\text{B.11})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2 (q_{x2} - qx1^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right). \quad (\text{B.12})$$

Using the identity

$$\mathcal{I} = \int_{-\infty}^{\infty} \exp(iax^2) dx = \frac{i\pi}{a} \quad (\text{B.13})$$

and labelling

$$a = \frac{k(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}} \quad (\text{B.14})$$

allows equation B.12 to be simplified to

$$\frac{1}{|q_{x1}| |q_{x2}|} \left(\int_{-\infty}^{\infty} \exp\left(\frac{ikx^2(q_{x2} - q_{x1}^*)}{2q_{x1}^* q_{x2}}\right) dx \times c.c. \right) = \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{i\pi(2q_{x1}^* q_{x2})}{k(q_{x2} - q_{x1}^*)}} \sqrt{\frac{-i\pi(2q_{x1} q_{x2}^*)}{k(q_{x2}^* - q_{x1})}} \quad (\text{B.15})$$

$$= \frac{1}{|q_{x1}| |q_{x2}|} \sqrt{\frac{4\pi |q_{x1}|^2 |q_{x2}|^2}{k^2 |q_{x2} - q_{x1}^*|^2}} \quad (\text{B.16})$$

$$= \frac{2\pi}{k} \frac{1}{|q_{x2} - q_{x1}^*|}. \quad (\text{B.17})$$

This means that the numerator of equation B.5 is

$$\frac{4\pi^2}{k^2} \frac{1}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{B.18})$$

Since E_x and E_y are orthogonal, $|E|^2 = |E_x|^2 |E_y|^2$, an expression for the denominator of equation B.5 can be found by:

$$\int_A |E|^2 dA = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |E_x|^2 |E_y|^2 dx dy \quad (\text{B.19})$$

$$= \int_{-\infty}^{\infty} \frac{1}{|q_x|} \exp\left(\frac{ikx^2(q_x - q_x^*)}{2q_x q_x^*}\right) \times y \text{ term.} \quad (\text{B.20})$$

Using a similar approach as in the calculation of the numerator terms:

$$\int_A |E|^2 dA = \frac{1}{|q_x|} \sqrt{\frac{i2\pi |q_x|^2}{k(q_x - q_x^*)}} \frac{1}{|q_y|} \sqrt{\frac{i2\pi |q_y|^2}{k(q_y - q_y^*)}} \quad (\text{B.21})$$

$$= -\frac{2\pi}{k} \frac{1}{\sqrt{(q_x - q_x^*)(q_y - q_y^*)}} \quad (\text{B.22})$$

we get the mode matching $\mathcal{M}$ to be

$$\mathcal{M} = \frac{\frac{4\pi}{k^2}}{-\frac{2\pi}{k} \times -\frac{2\pi}{k}} \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)} \sqrt{(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|} \quad (\text{B.23})$$

$$= \frac{\sqrt{(q_{x1} - q_{x1}^*)(q_{y1} - q_{y1}^*)(q_{x2} - q_{x2}^*)(q_{y2} - q_{y2}^*)}}{|q_{x2} - q_{x1}^*| |q_{y2} - q_{y1}^*|}. \quad (\text{B.24})$$

B.2 Effect of Error in the Radius of Curvature of a Mirror on the defocus of a Beam

The error in the optical power of a mirror due to an error in the radius of curvature of is

$$S = \frac{2}{R_m} \quad (\text{B.25})$$

$$dS = \frac{-2}{R_m^2} dR. \quad (\text{B.26})$$

The error in the beams defocus is the same as the error in the radius of curvature of the the mirror. The effect this has on the beam parameter q can be found by

defining a contour of modes distance dq from q i.e. $q + |dq| e^{i\theta}$, where dq is

$$q = \frac{q_i}{Sq_i + 1} \quad (\text{B.27})$$

$$dq = \frac{q_i^2}{(-Sq_i + 1)^2} dS. \quad (\text{B.28})$$

This gives you a family of modes which will contain the modes due to an error in the radius of curvature of the mirror, but since q is a complex number the contour you need to define to encapsulate the concept of an error in q also contains modes which are have different beam sizes as well as different defocuses. To do the calculation this way is cumbersome, it is much more straight forward to compute the effect of having a mirror ROC error by evaluating at every mirror

$$\begin{pmatrix} 1 & 0 \\ \pm dS & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ -S & 1 \end{pmatrix} \begin{pmatrix} q \\ 1 \end{pmatrix} = \begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix} \quad (\text{B.29})$$

and then divide the top and bottom entries of the $\begin{pmatrix} q \\ (S \pm dS)q + 1 \end{pmatrix}$ vector.

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